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Organization of the
United Nations



ICARDA
Science for resilient livelihoods in dry areas



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WEBINARS SERIES ON RAINWATER HARVESTING

27th June 2022

Module n°3:
Mapping the potential for
rainwater harvesting

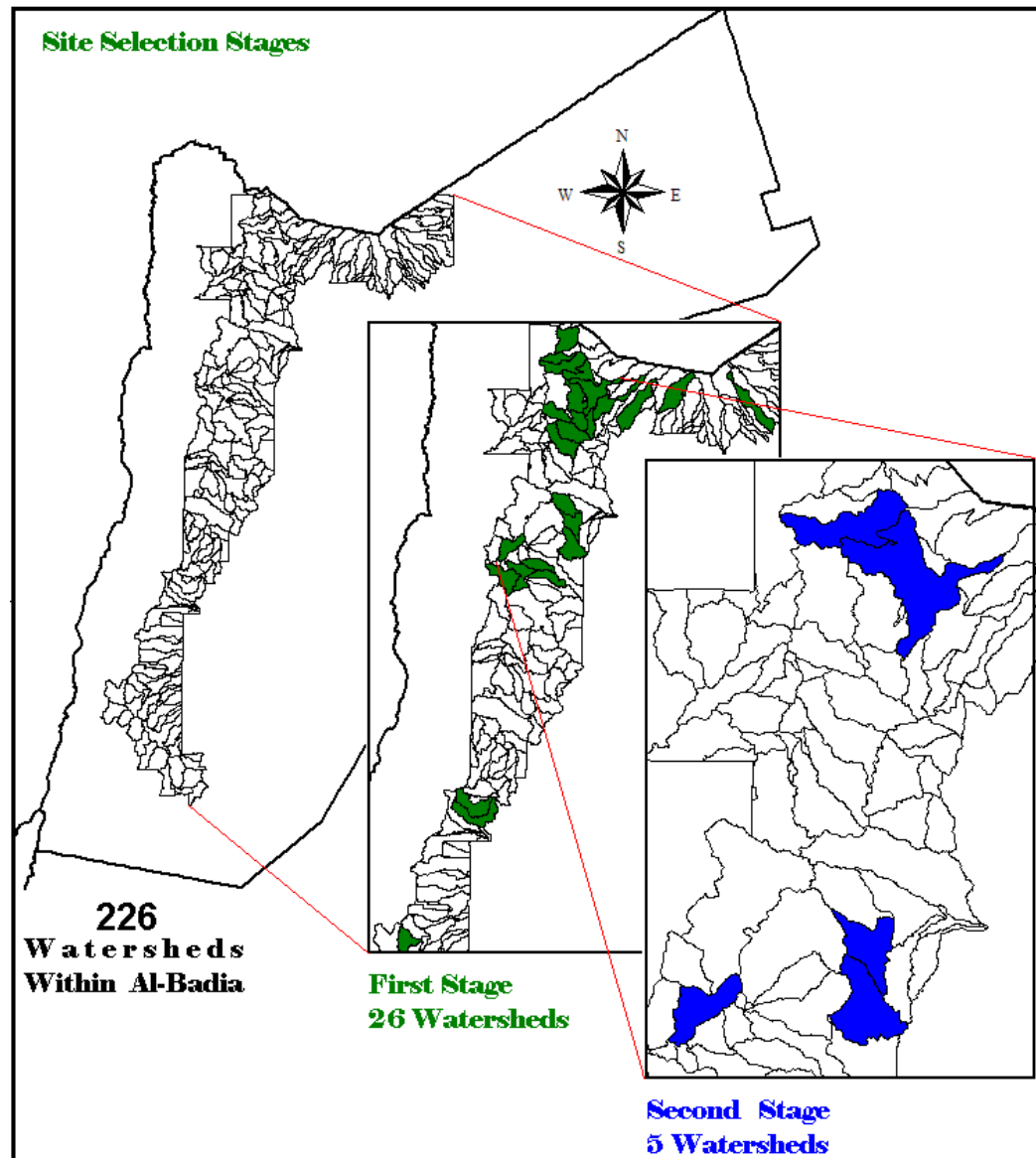


A Participatory GIS Approach for Assessing Land Suitability for Rainwater Harvesting in an Arid Rangeland Environment

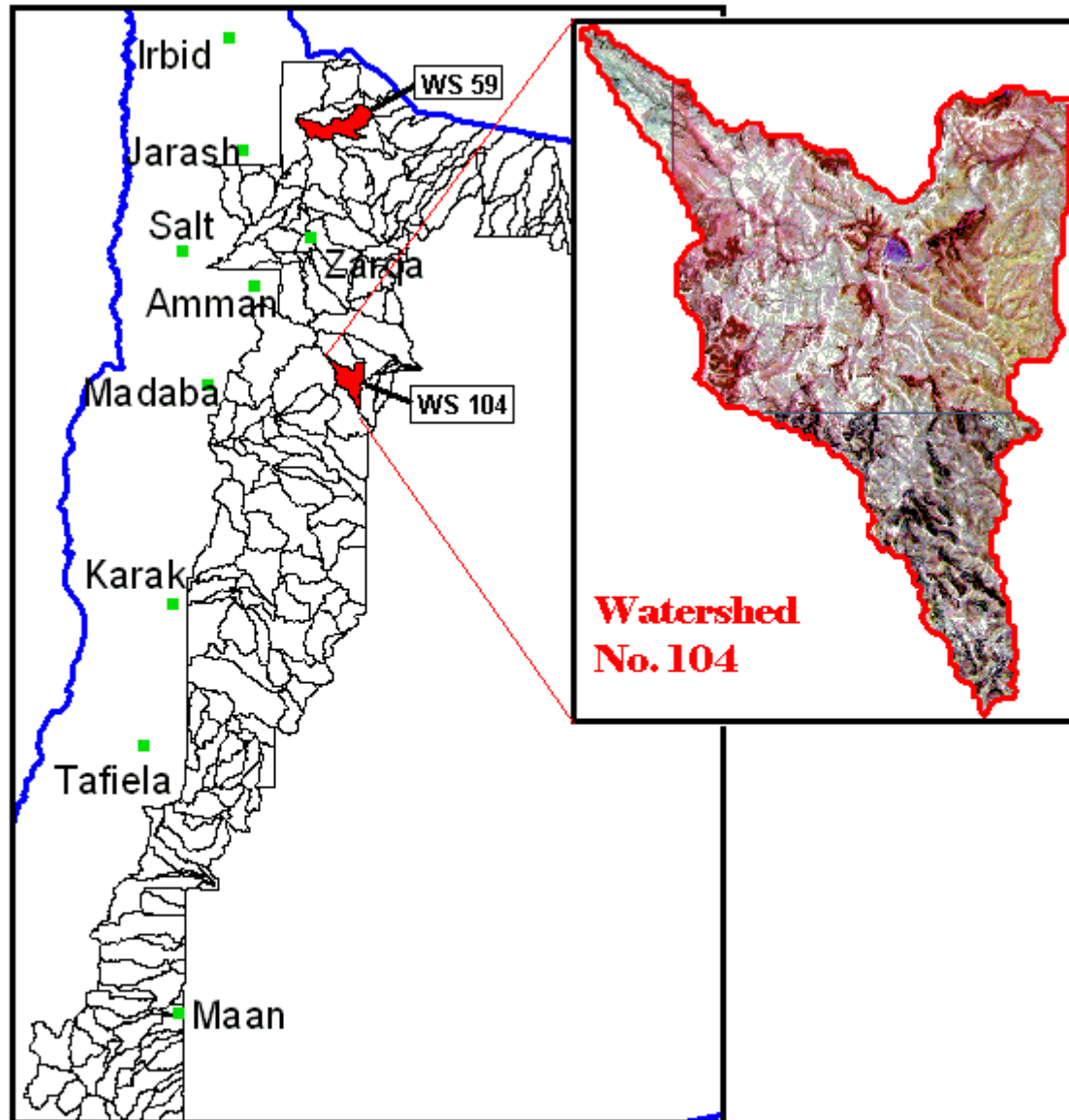
Feras Ziadat
FAO



Site Selection



Site Selection



Stage Three
2 Watersheds

Muhareb Watershed

- 60 km²
- 100-150 mm precipitation
- Silty clay loam soils, erosive
- Gentle slopes (4%)
- Barley, rangeland

Objective of watershed characterization

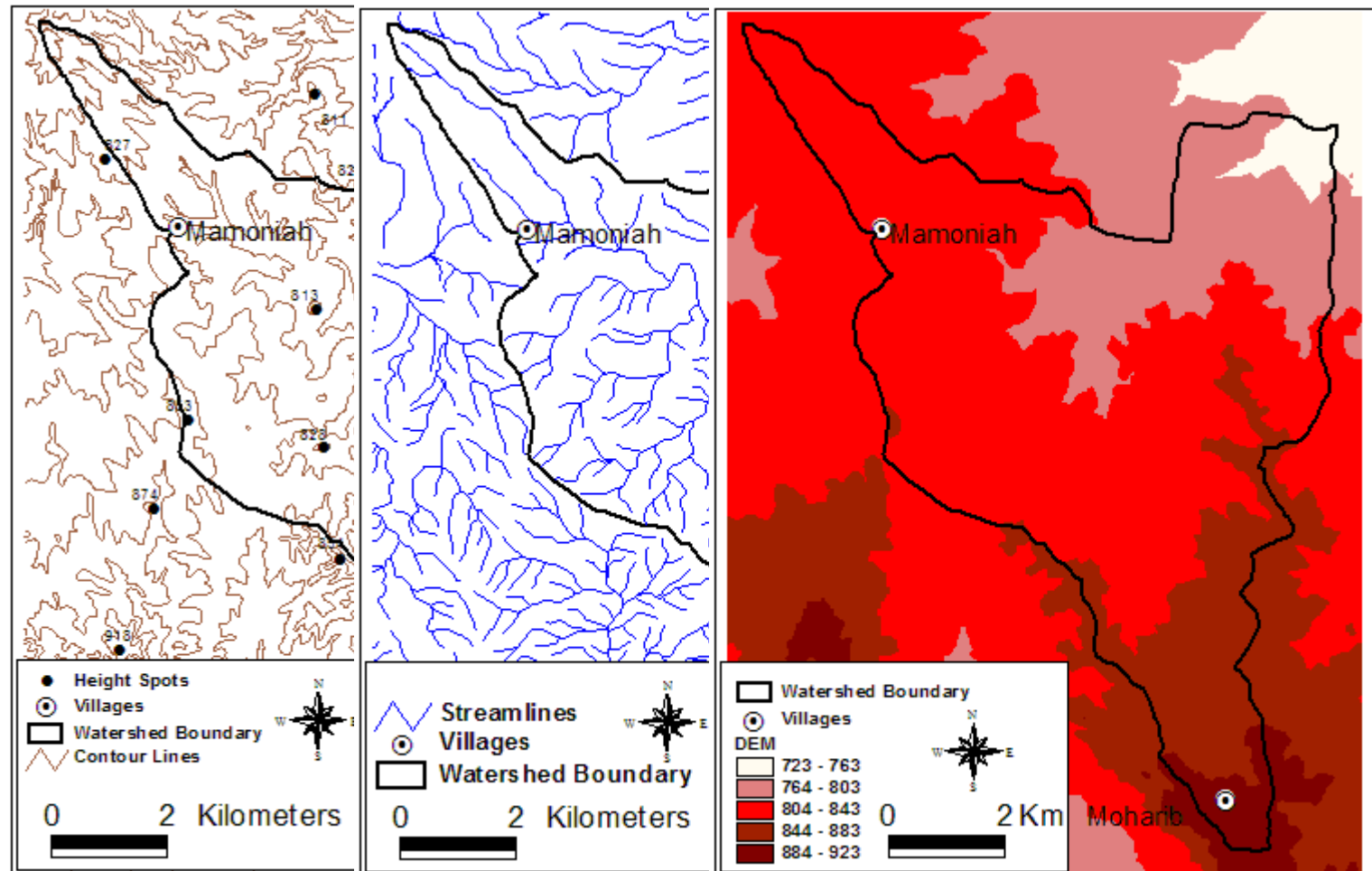
- To develop a methodology to assess in a systematic, practical, and informed manner the potential for water harvesting in watersheds in arid and semi-arid regions.

Watershed Characterization

DATA COLLECTION

1. AVIALABLE DATA

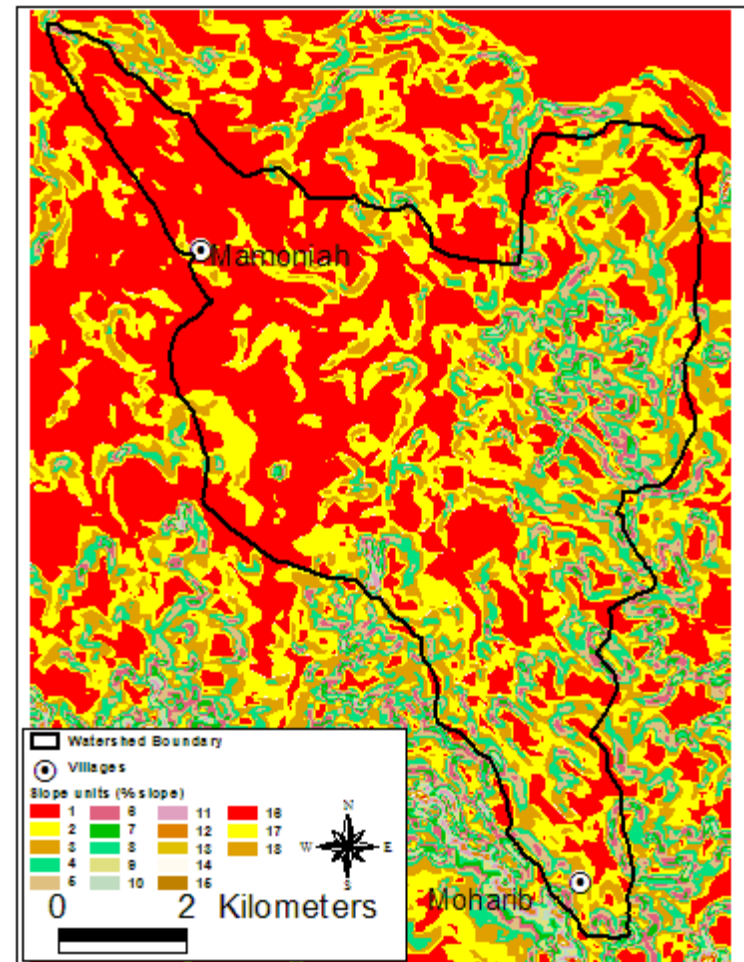
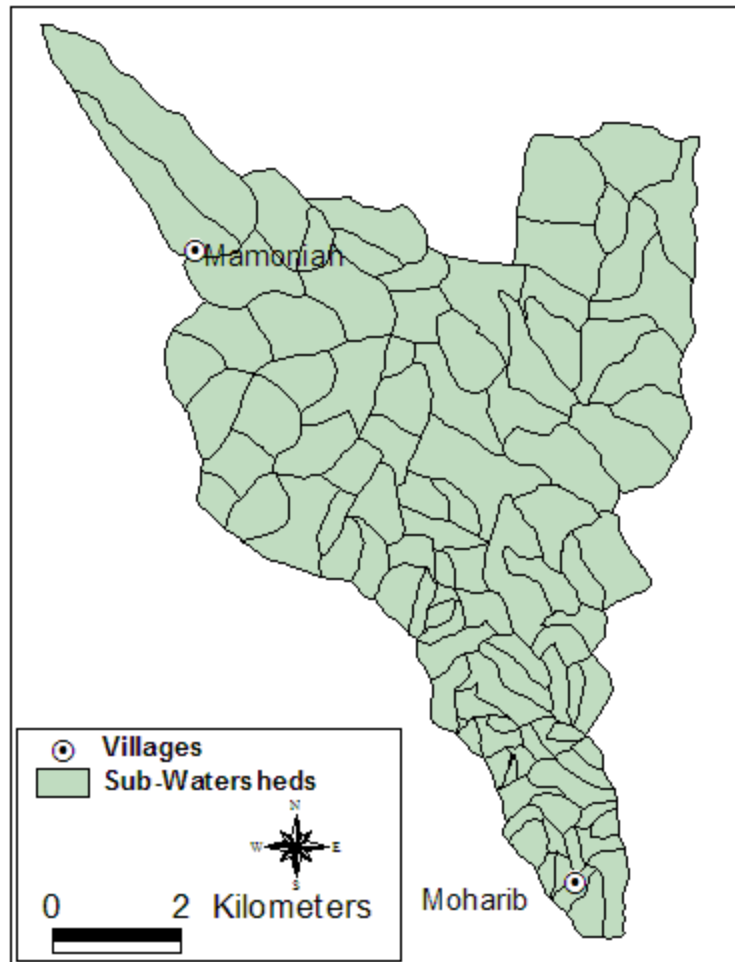
Data from the Royal Jordanian Geographic Center (RJGC)



DATA COLLECTION

1. AVIALABLE DATA

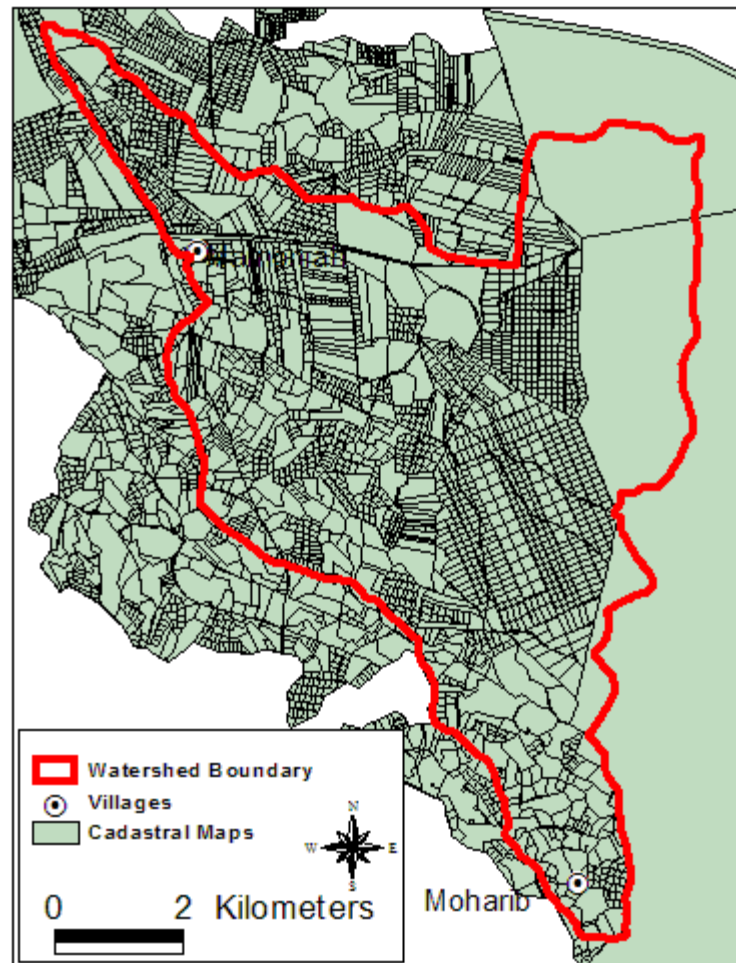
Data derived from topographic information



DATA COLLECTION

1. AVIALABLE DATA

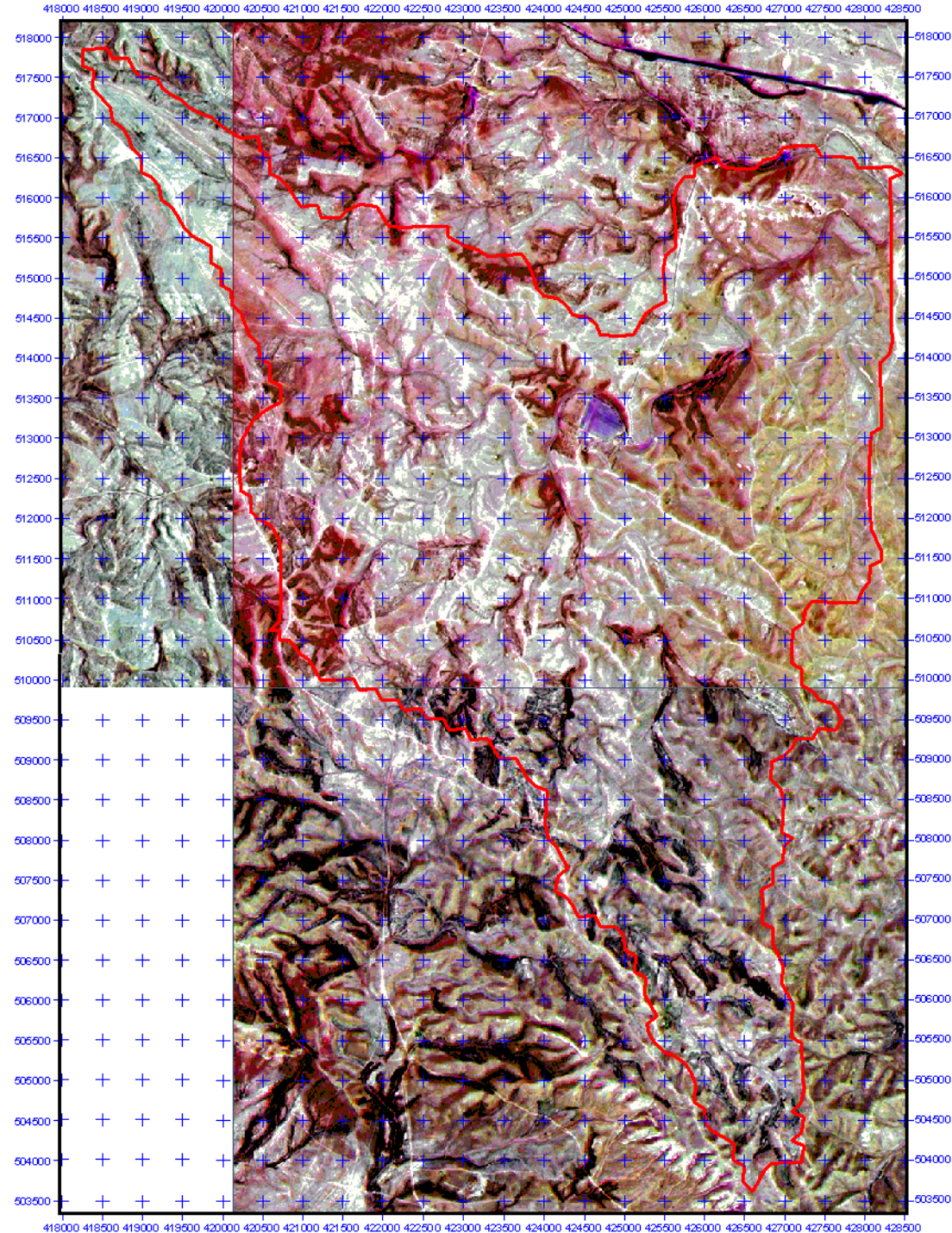
Data from the Department of Land and Surveying (DLS)



DATA COLLECTION

2. DATA COLLECTED FROM THE FIELD

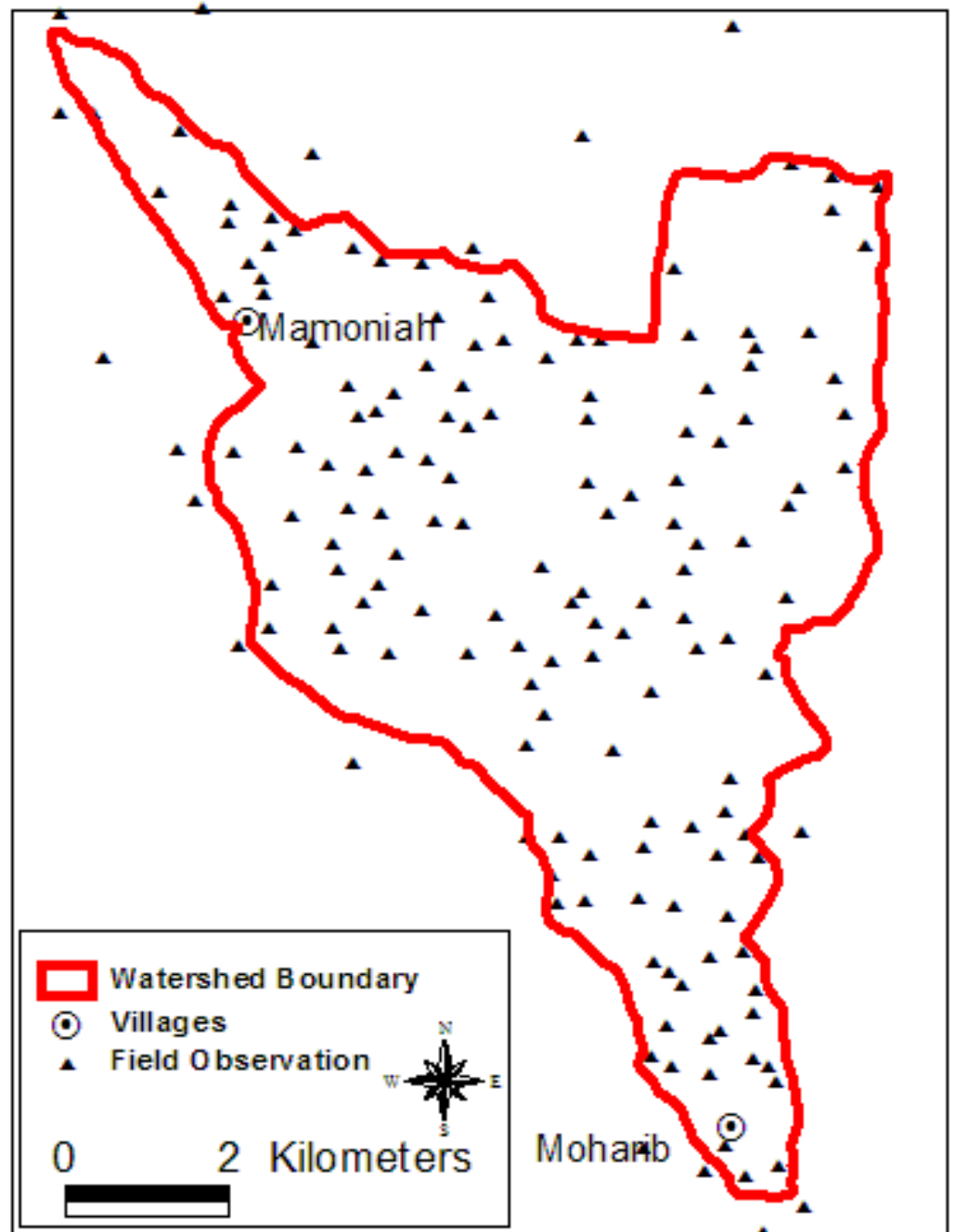
Sampling procedure



DATA COLLECTION

2. DATA COLLECTED FROM THE FIELD

Field Observations



DATA COLLECTION

2. DATA COLLECTED FROM THE FIELD

Attributes recorded

1. Slope Steepness (percent): use Abbney level.
2. Surface cover of the land (stoniness percent).
3. Vegetation type and coverage (percent, for some sites).
4. Texture of the surface horizon: estimated by feeling.
5. Limiting soil depth (cm, depth to limiting feature).
6. GPS coordinates: easting, northing and elevation.

SUITABILITY FOR WATER HARVESTING INTERVENTIONS

1. REQUIREMENTS FOR WATER HARVESTING

Technique	Crop	Soil				Land Slope (2)		Land Cover				Socio-economic	
		Depth (1)		Texture				Vegetation (3)		Stoniness (4)		Farm Size (5)	
		P1	P2	P1	P2	P1	P2	P1	P2	P1	P2	P1	P2
Micro-catchments On-farm systems													
Contour ridges	Range	shl	med	med	var	med	steep	poor	med	med	low	var	var
	Field	med	deep	med	var	low	med	poor	poor	low	low	sml	med
	Trees	deep	deep	heavy	med	med	low	poor	poor	low	med	sml	med
Semi-circular and triangular bunds	Range	shl	med	var	var	med	low	poor	med	med	low	var	var
	Field	med	deep	med	heavy	low	med	poor	poor	low	low	sml	med
Small basins	Trees	deep	deep	heavy	med	low	low	poor	med	low	med	sml	sml
Runoff strips	Range	med	var	med	var	med	low	poor	med	med	var	med	var
	Field	deep	med	med	med	low	med	poor	poor	low	med	sml	med
Inter-row system	Trees	deep	deep	heavy	med	low	low	poor	med	low	med	sml	med
Contour bench terraces	Trees	deep	med	heavy	heavy	steep	med	poor	med	low	med	sml	med
	Field	deep	med	heavy	var	steep	med	poor	poor	low	med	sml	med
Narrow-base Contour terraces (Gradoni)	Trees	deep	deep	med	heavy	steep	med	poor	med	low	low	sml	med
	Range	deep	med	med	var	med	steep	med	low	low	med	med	var
Macro-catchments Wadi-bed systems													
Trapezoidal bunds (Cultivated area)	Field	med	med	heavy	med	low	med	poor	poor	low	med	sml	med
Catchment area	-	shl	var	var	heavy	med	steep	poor	poor	low	med	large	med
Wadi – bed cultivation (Water spreading)	All crops	deep	med	med	heavy	low	low	low	med	Var	var	med	sml
Jessour	Trees	med	var	heavy	med	med	steep	var	med	med	var	sml	med
Off – wadi systems													
Water spreading (diversion)	Field	med	deep	med	heavy	low	med	poor	med	low	med	med	var
	Trees	deep	med	med	heavy	low	low	poor	poor	med	low	med	sml
Large bunds	Trees	deep	deep	heavy	med	low	low	poor	var	med	low	med	large
	Field	med	var	med	var	low	low	poor	poor	low	med	med	med
	Range	med	var	var	var	med	low	var	poor	var	var	med	large
Hillside runoff systems (cultivated)	Field	med	med	med	var	low	med	poor	med	low	var	med	var
	Trees	deep	med	med	heavy	low	med	poor	med	low	var	med	med
Catchment	-	shl	var	var	heavy	steep	med	poor	poor	low	med	large	med
Tanks and Hafirs	All crops	Variable		med	heavy	low	var	poor	med	variable		med	var
Cisterns	-	shl	med	variable		variable		poor	med	variable		sml	med

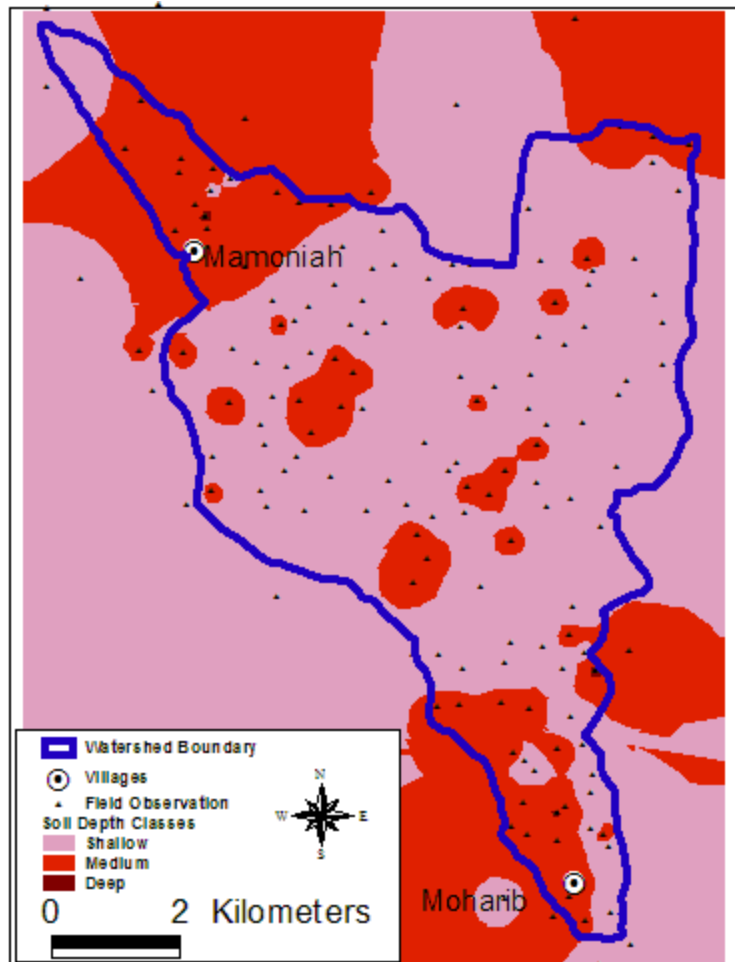
(1) shl < 50 cm, med: 50-100 cm, deep > 100 cm, (2) low < 4%, med: 4-12%, steep > 12%, (3) poor < 15%, med: 15-30%, dense > 30%,

(4) low < 10%, med: 10-25%, high > 25%, (5) sml < 5 ha, med: 5-25 ha, large > 25 ha

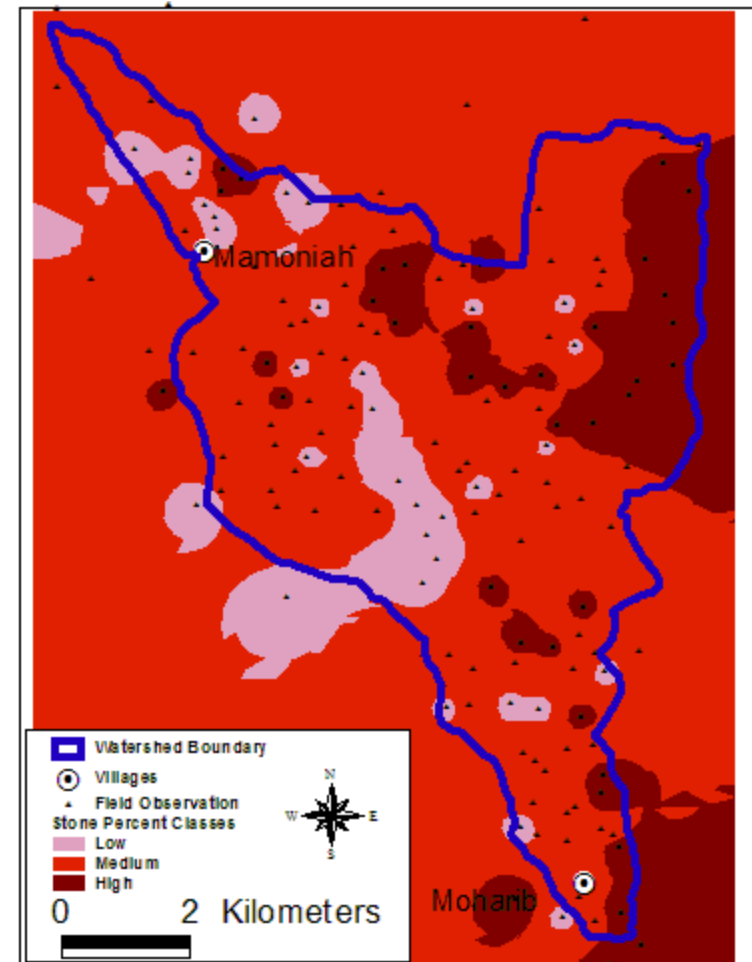
SUITABILITY FOR WATER HARVESTING INTERVENTIONS

2. PHYSICAL CONDITIONS OF THE WATERSHED

Limiting Soil Depth

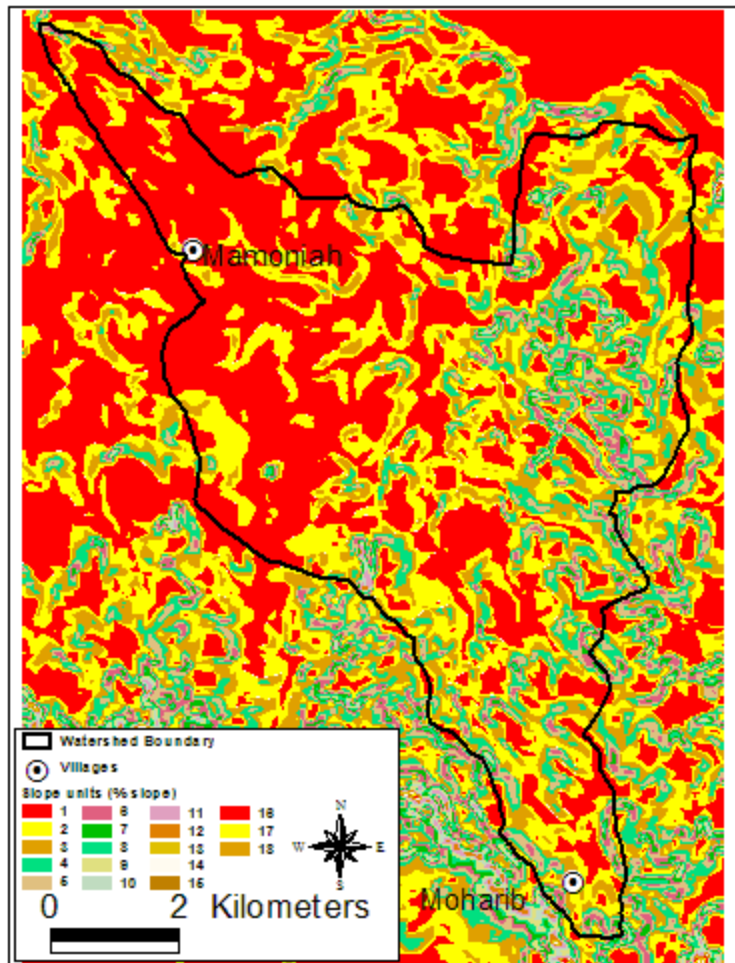


Stone Percent

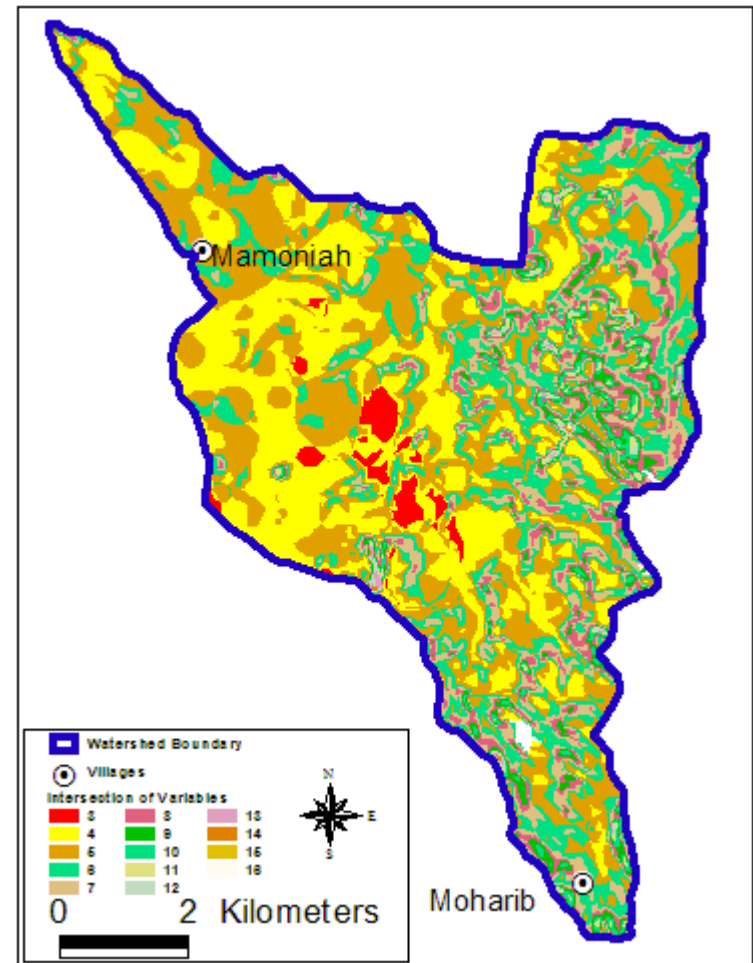


SUITABILITY FOR WATER HARVESTING INTERVENTIONS

3. LAYERS OVERLAY AND SUITABILITY ANALYSIS



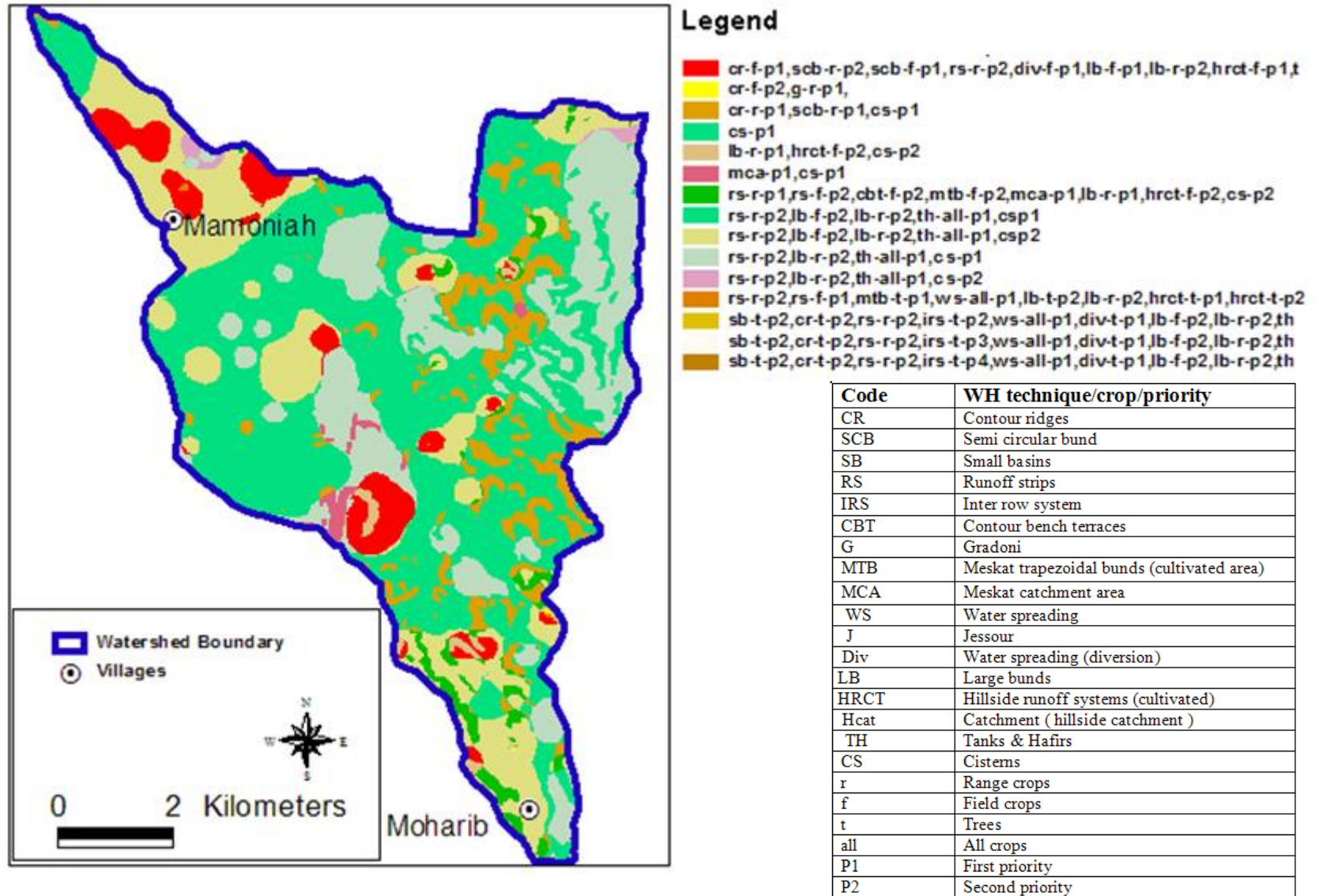
Slope Classes



Physical conditions of Slope Classes

SUITABILITY FOR WATER HARVESTING INTERVENTIONS

3. LAYERS OVERLAY AND SUITABILITY ANALYSIS



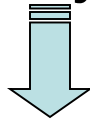
Land suitability for water harvesting

Final selection and implementation

Integrated approach

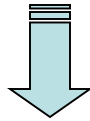
Suitability maps

Bio-physically suitable



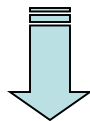
Field visits

Interdisciplinary team discussion

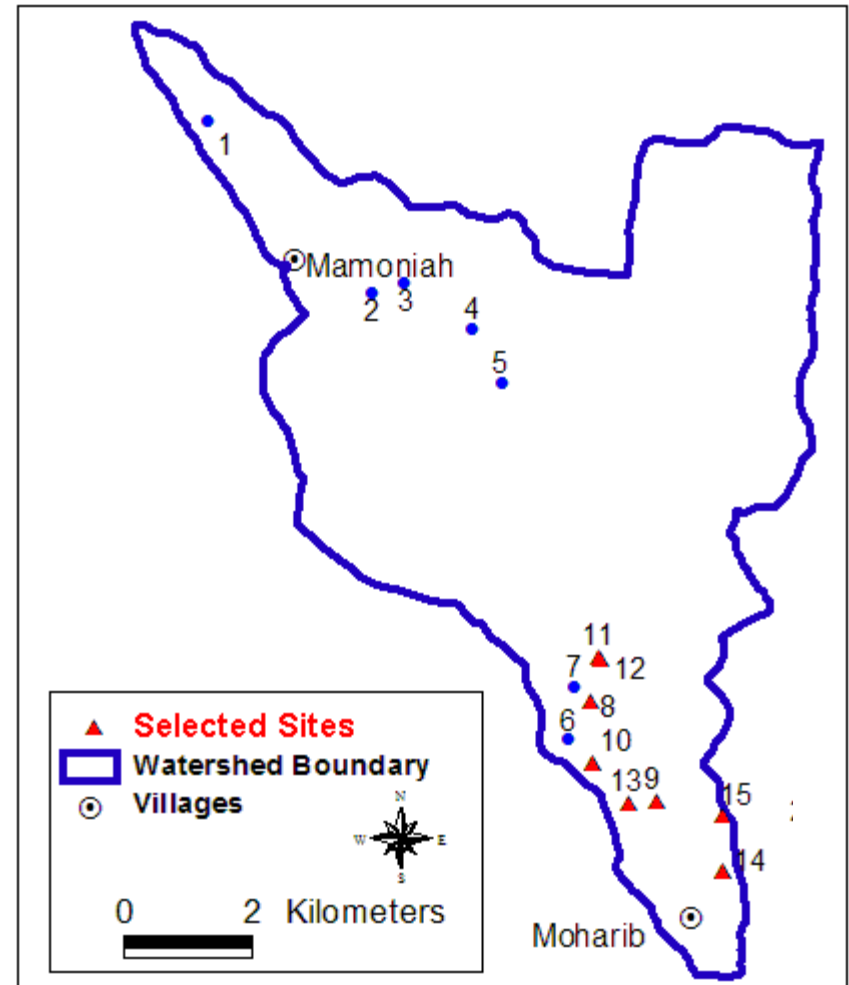


Land tenure

Socio-economically sound options

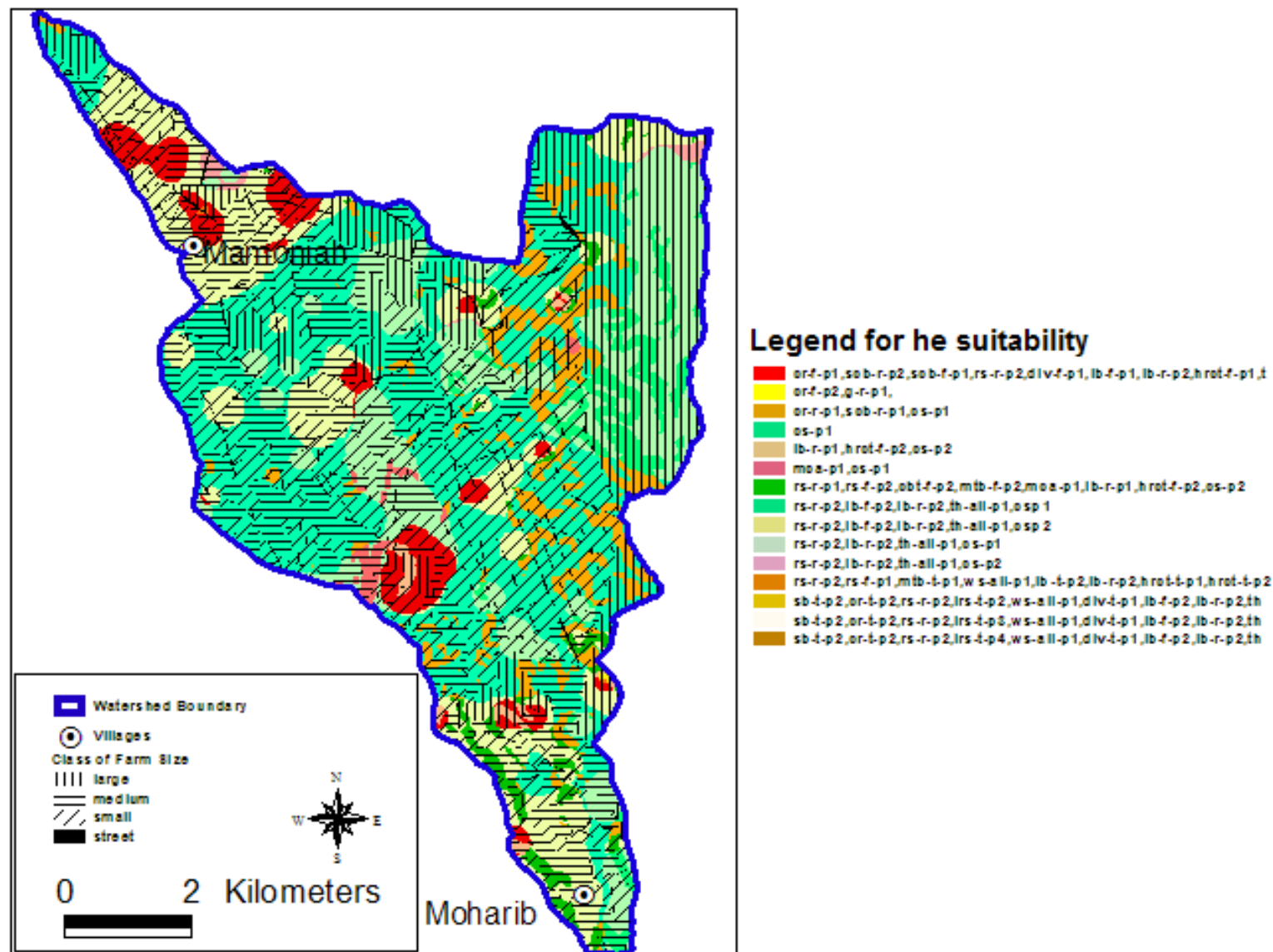


**Implementation
& Monitoring**



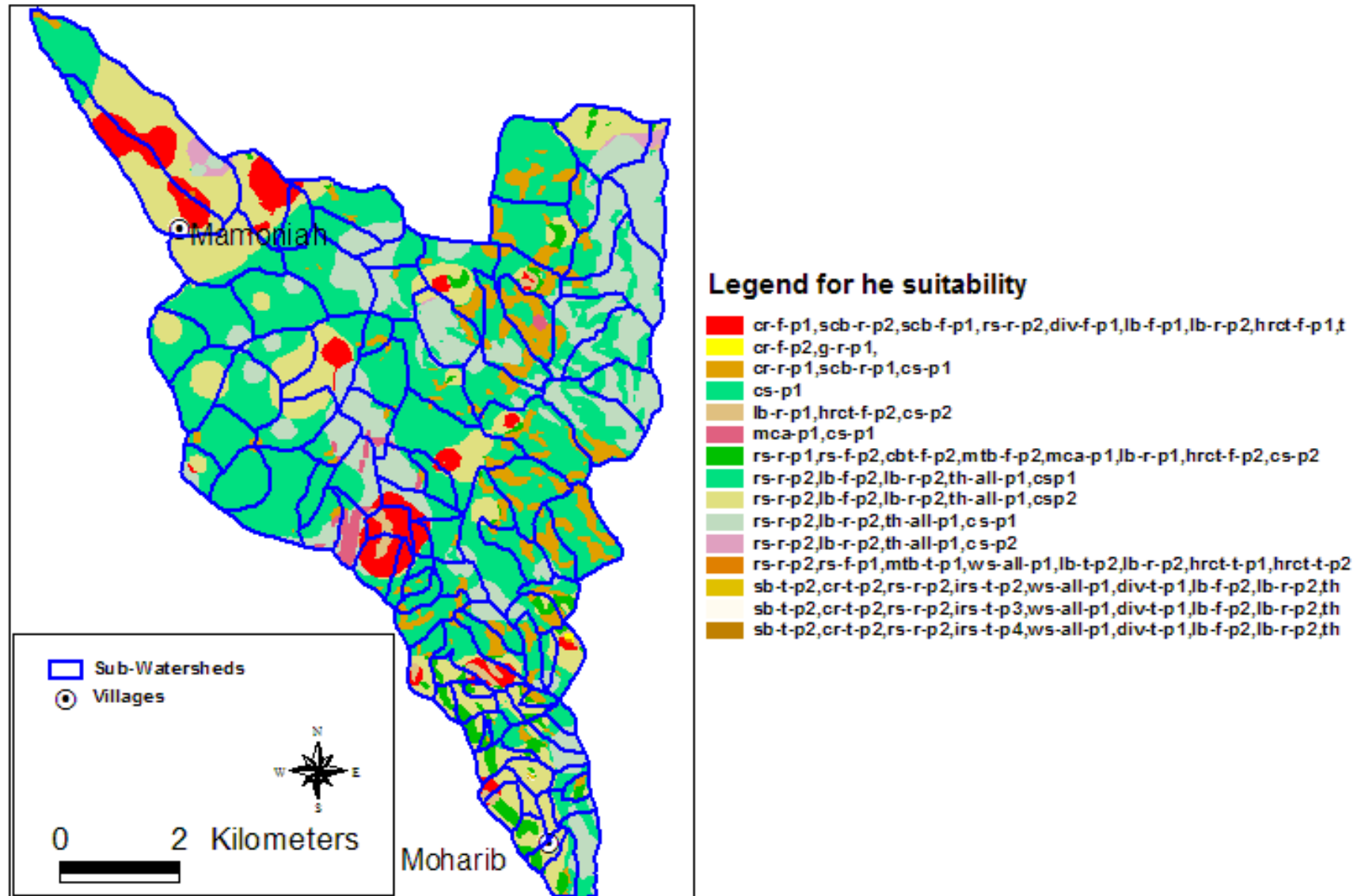
UTILITY OF THE SUITABILITY RESULTS

1. FARM SIZE AS SELECTION CRITERIA



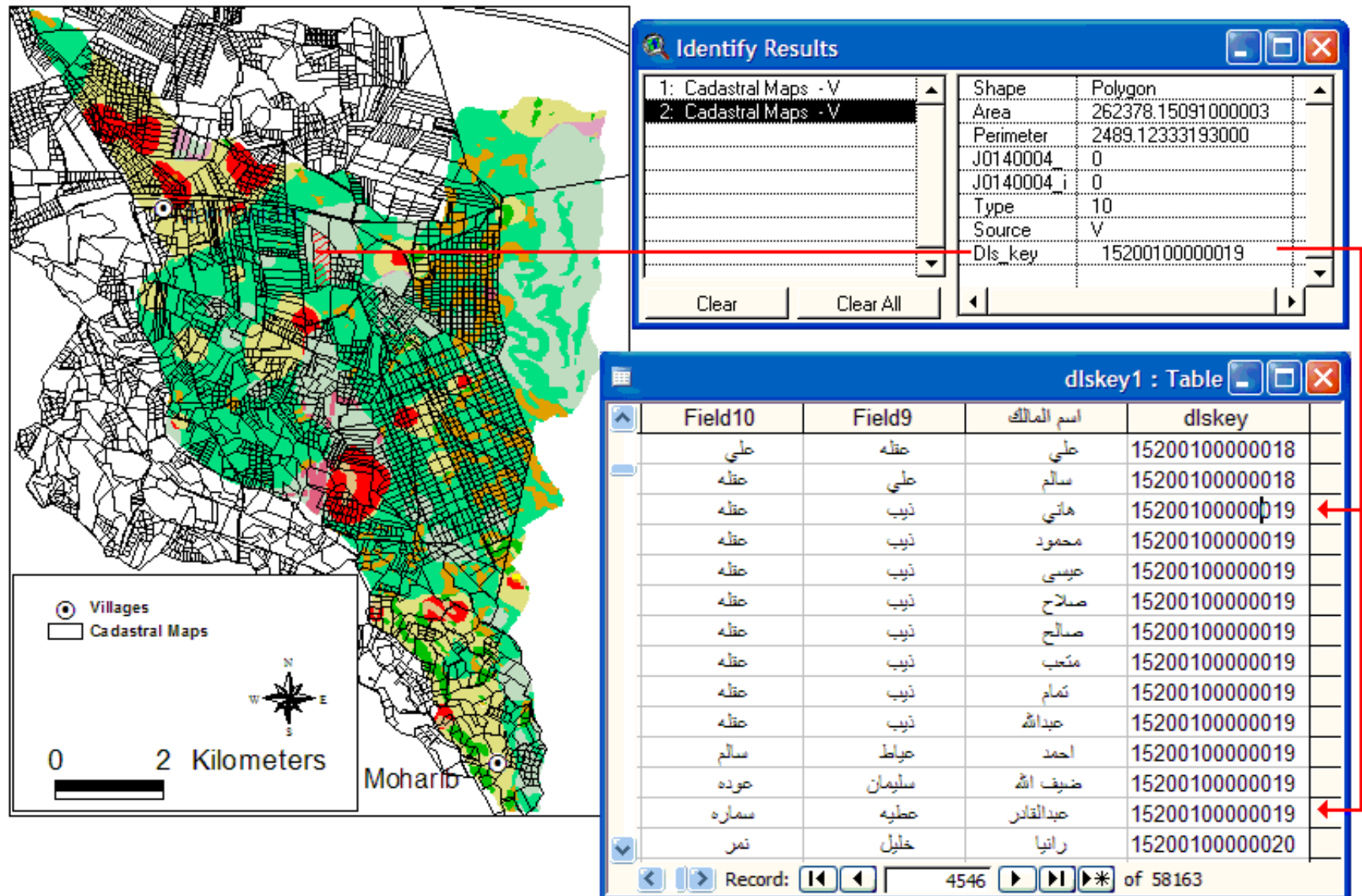
UTILITY OF THE SUITABILITY RESULTS

2 SUITABILITY AND WATERSHED SUBDIVISION



UTILITY OF THE SUITABILITY RESULTS

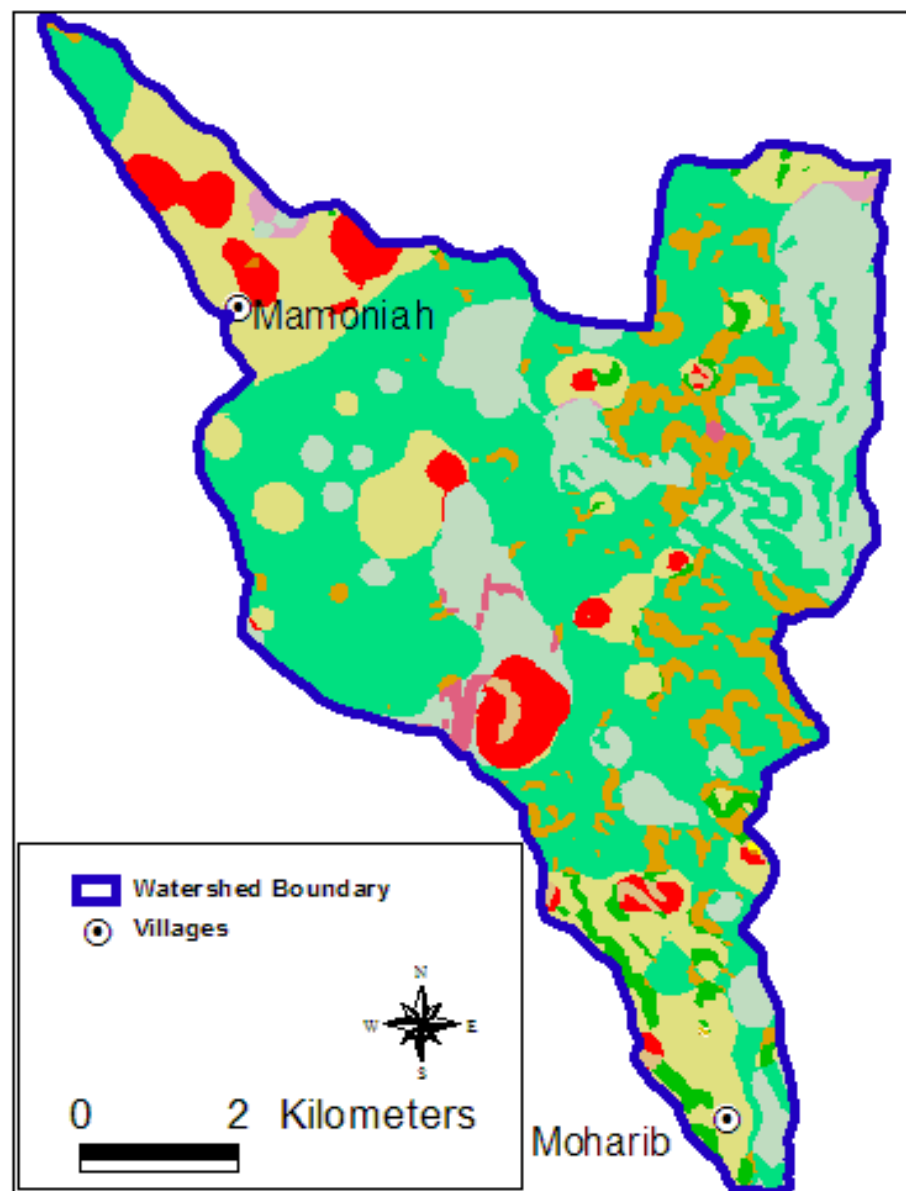
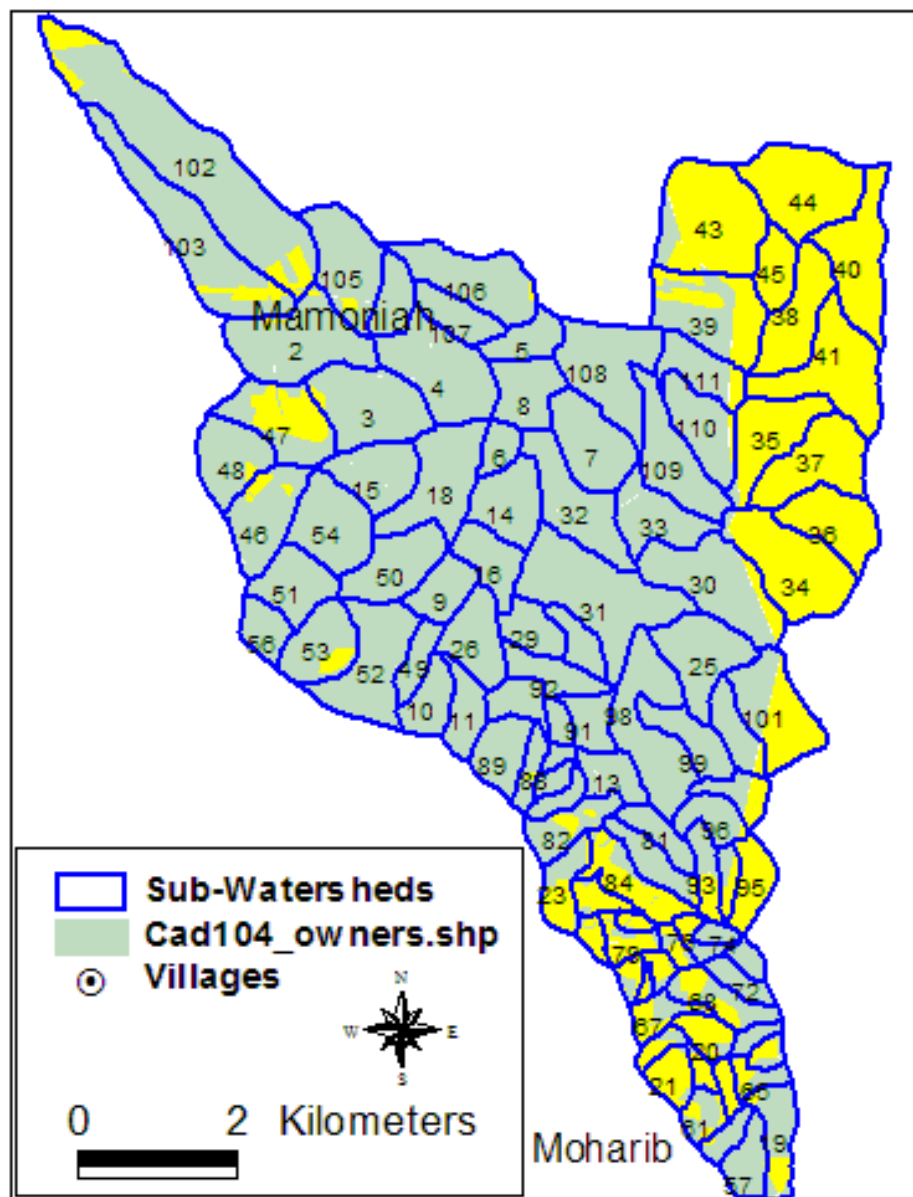
3 SUITABILITY AND LAND OWNERS



UTILITY OF THE SUITABILITY RESULTS

3 SUITABILITY AND LAND OWNERS

Local (indigenous) owners



USING THE SUITABILITY ANALYSES RESULTS FOR THE IMPLEMENTATION STAGE

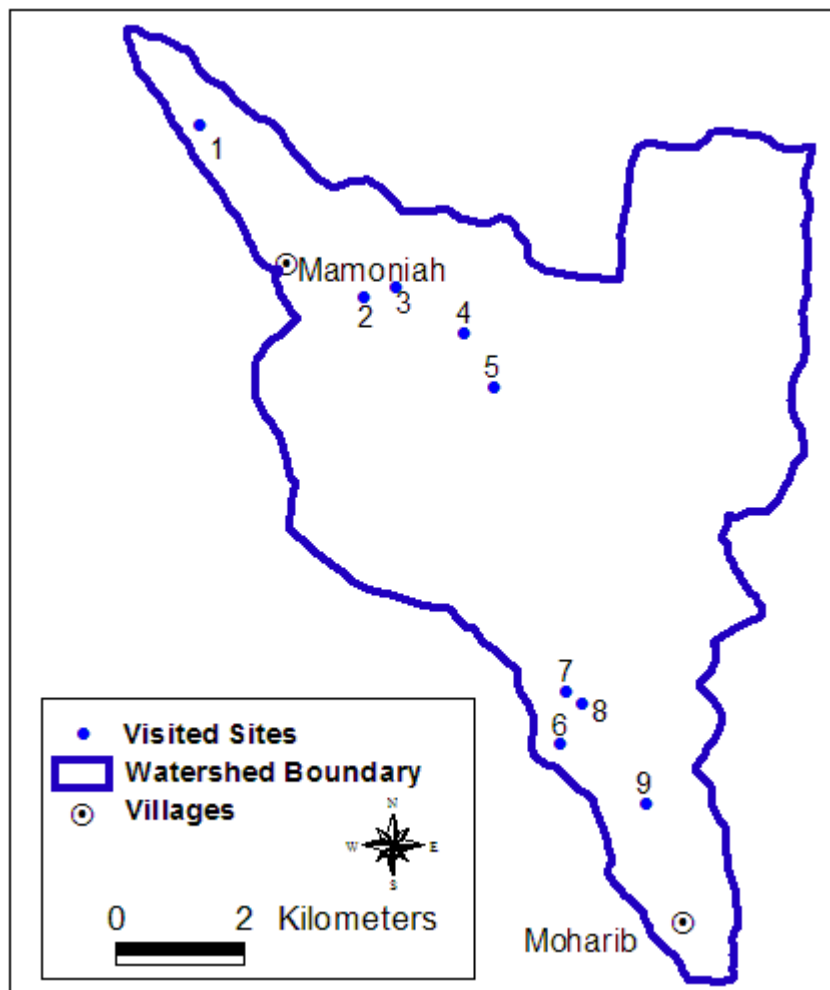
IDENTIFICATION OF SUITABLE LOCATIONS FOR WH INTERVENTIONS

Field Visits by multi-disciplinary team

Data used:

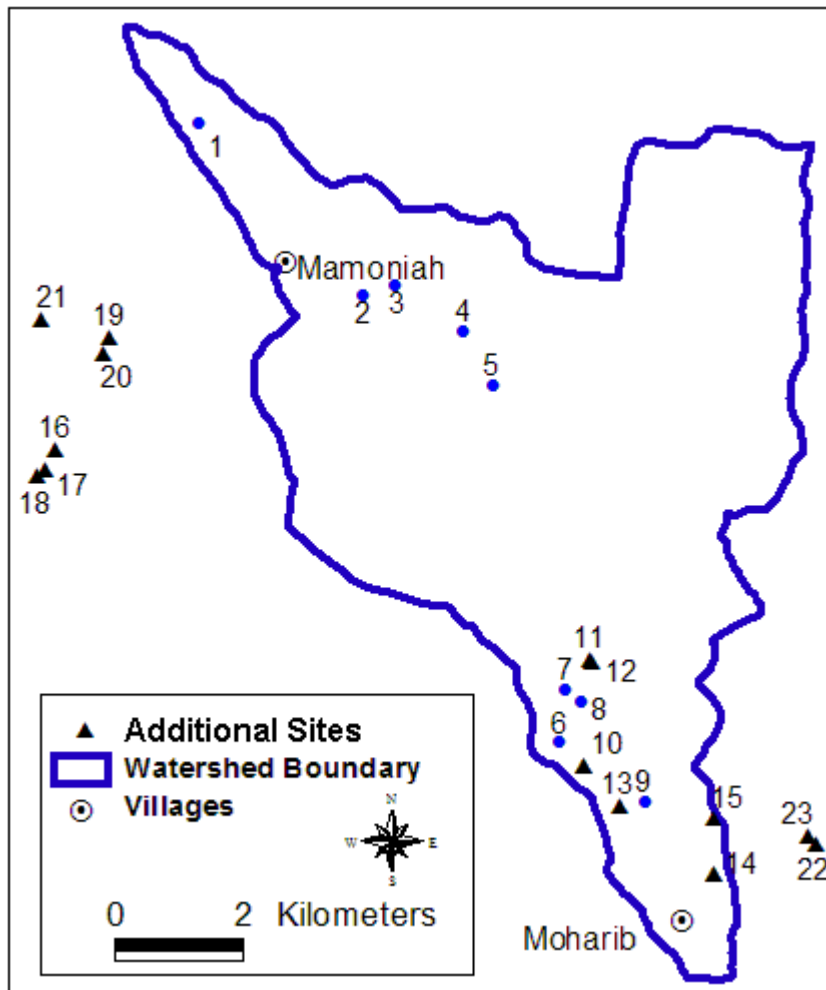
- 1. Land suitability for different WH techniques**
- 2. Locations of potential earth dams and Hafaier (from the hydrological analysis)**
- 3. Satellite images and GPS (used for navigation)**
- 4. Cadastral maps**

Selected sites (first field visit)



Site No.	Proposed Intervention
1	- On-Farm WH*
2	- Check dam
3	- First option: Pond (Hafaire) - Second option: Check dam
4	- Pond (Hafaire)
5	- Earth dam
6	- Earth dam
7	- Water Spreading
8	- Check dam
9	- On-Farm WH

Selected sites (after discussions with farmers)



Site No.	Proposed Intervention
10	- Check dam
11	- Pond (Hafaire)
12	- Pond (Hafaire)
13	- On-Farm WH*
14	- On-Farm WH
15	- On-Farm WH
16	- On-Farm WH
17	- On-Farm WH
18	- On-Farm WH
19	- Water Spreading
20	- On-Farm WH
21	- Pond (Hafaire)
22	- Pond (Hafaire)
23	- Earth dam

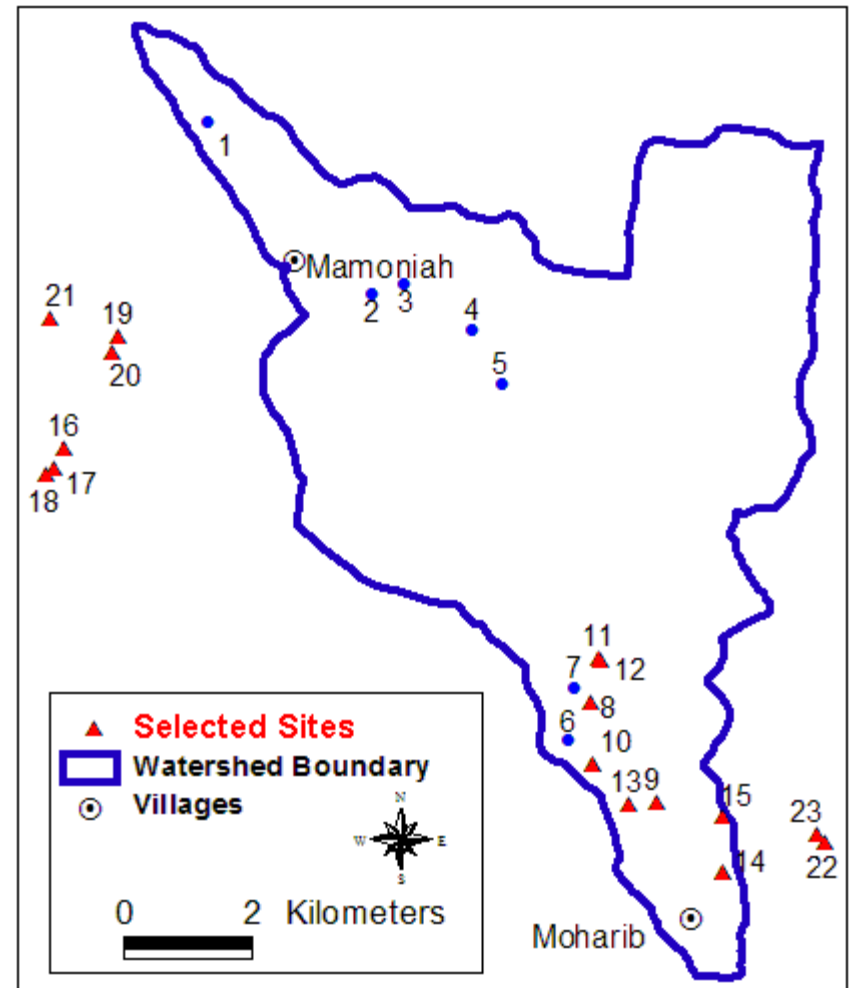
Final Selection:

Biophysically Suitable

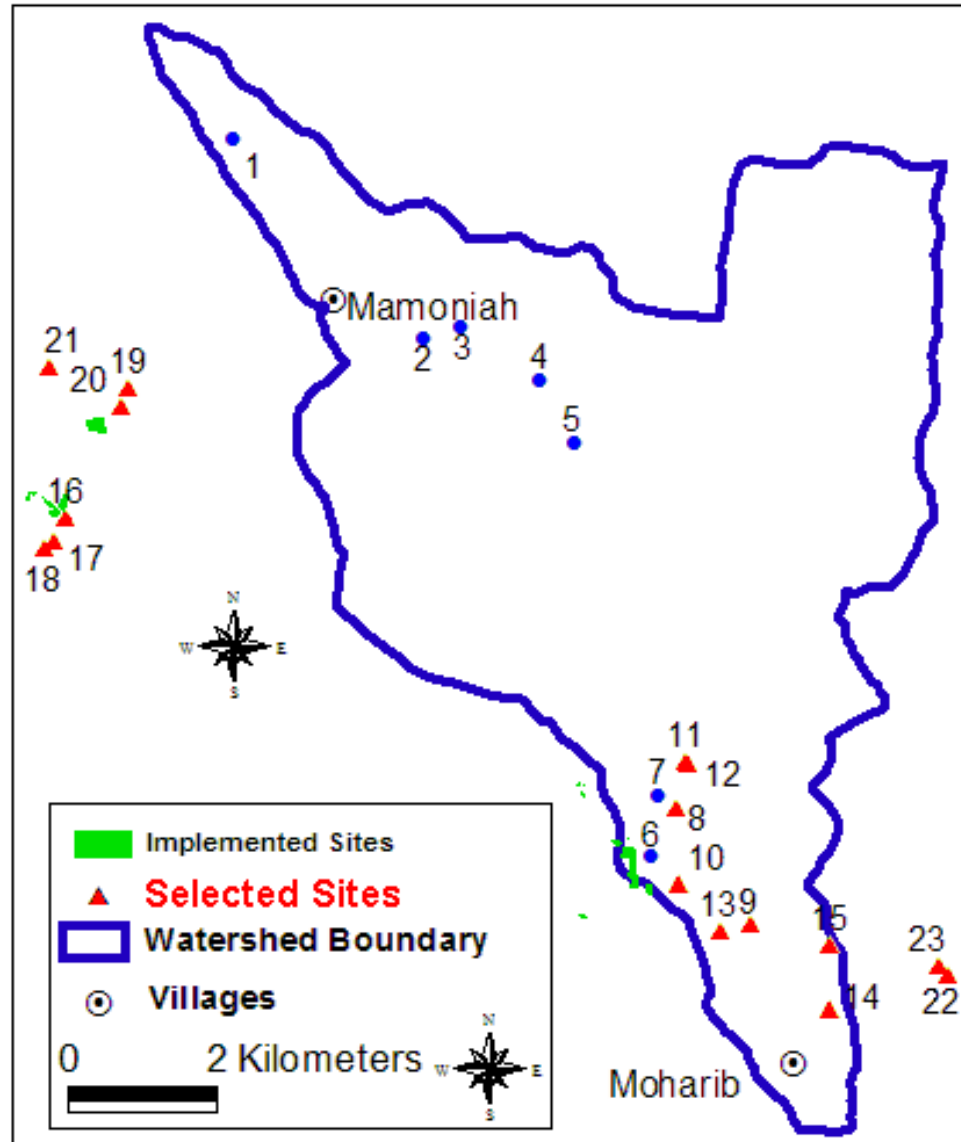
Socio-economically acceptable



Field Work Plan



Implemented Sites



Concluding remarks

- A systematic approach for matching water harvesting requirements with physical conditions of the land was developed
- GIS facilitates an iterative suitability analyses and integration of biophysical and socio-economic aspects
- The developed suitability map helped to identify fields with farmers where water-harvesting techniques were implemented





Thank you
for your
attention

